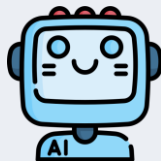
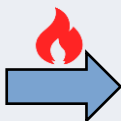


Previous Methods
(Task $\rightarrow$ Trajectory)

Click



Type

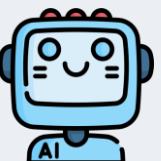
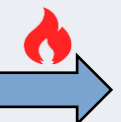


Atomic Task Training

Small MLLM

Poor Compositional Gen.

Task: Find me a
blue snowboard
with 4+ stars.



Wrong distilled
trajectory:
yellow snowboard
with 3 stars

Coarse Task Training

Small MLLM

Misaligned Trajectory &
Weak Constraints

Our Method: PEEU
(Task $\rightarrow$ Trajectory $\rightarrow$ Experience $\rightarrow$ Task)

Task: Find me a
blue snowboard
with 4+ stars.



★★★★

Web Agent

click, site



type, snowboard

click, 3 stars



★★★

Interact



Chrome

Exploration Tree



Hindsight
Procedural
Experience

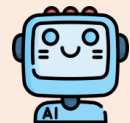


Turn trash
into treasure!

PEEU Task
(Aligned):
Find me a
yellow
snowboard
with 3 stars.



High Quality



Small MLLM

Step 1: Agent Opens
shopping site.

Step 2: Agent
Searches snowboard.

Step 3: Agent clicks a
product with **3 stars**.

Step 4: Agent clicks a
product in **yellow color**.

Step 5: Agent Clicks
product link.